SolutionRequirements

Date	14 July 2026
Team ID	
Project Name	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements

This document defines the functional and non-functional requirements for the **OptiCrop – Smart Agricultural Production Optimization Engine**, an AI-based crop recommendation system that predicts suitable crops using soil nutrients and environmental conditions.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement	Priority
1	User Interface	Provide a responsive web interface for users.	High
2	Input Validation	Validate soil and environmental input values.	High
3	Prediction	Predict the most suitable crop using the ML model.	High
4	Crop Information	Display crop details, image, and farming tips.	High
5	Navigation	Support Home, About, Find Crop, and Result pages.	Medium
6	Data Management	Process and retrieve crop information efficiently.	Medium

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Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Acceptance Criteria
1	Performance	Fast crop prediction.	< 5 seconds
2	Reliability	Accurate recommendations.	High accuracy
3	Availability	System available whenever required.	99% uptime
4	Usability	Simple and responsive interface.	User-friendly
5	Security	Validate user inputs.	Secure input handling
6	Maintainability	Easy to update and extend.	Modular design